Predictive Building Flood-Exposure Assessment for the Lower Kelani Basin

Component M1 — Progress 01 Research Report: data, labels, model comparison, hypothesis testing and demo

Project J26-DS-360 · AI-Powered Flood Disaster Impact Assessment and Decision Support System for Sri Lanka
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One-paragraph summary. Using free satellite data in Google Earth Engine, we built a building-level dataset for 341,971 buildings in the lower Kelani basin and labelled each one by its **flood exposure** during Cyclone Ditwah (distance to flood water detected by Sentinel-1 radar on 2 December 2025). Using only information available **before** the flood (Sentinel-2 imagery, elevation, HAND, river distance, building density), we trained and compared 10 models — classic machine learning and deep learning, with and without pre-training, including the proposal's Dual-Stream ResNet baseline and a first prototype of the proposed HSC-Attention model — under the same 10-fold **spatial** cross-validation. A Friedman test showed the models do not perform equally (p = 3.40e-13). The selected model is **D6_HSC_Attention** (macro-F1 0.666 ± 0.030). We then **optimised** the architecture and training settings of HSC-Attention and of Baseline 1 with nested spatial cross-validation (Section 9): optimised HSC-Attention reached macro-F1 0.676 ± 0.037 (default 0.660; gain not significant after Holm correction). Under the same optimisation budget, HSC-Attention beat the Dual-Stream baseline in 10 of 10 folds (Holm p = 0.012), and ablations show that cross-attention, the DEM-derived terrain bands and the terrain stream each contribute significantly. An interactive local web demo shows predictions for every building.

Honesty note for the panel. These are measured results, not the proposal targets. The task was adapted from "damage grade" to "flood exposure" because no building-damage labels exist for Sri Lanka and Sentinel-2 (10 m) cannot see roof damage. Damage-percentage regression (MAE) and segmentation IoU are therefore **not** reported, because valid ground truth does not exist for them.

1. What the whole project does — step by step

This section is the "map" of the research. Each step lists what was done, why, and its status. Steps marked **DONE** were executed and verified; **NEXT** are planned after Progress 01.

#	Step	What we did / will do (simple words)	Status
1	Proposal review	Read the full proposal, found risks (no damage labels for Sri Lanka, MAE/IoU not measurable, unverifiable claims).	DONE
2	Environment	Laptop has no NVIDIA GPU, so all training ran on free Google Colab (Tesla T4 GPU). GEE project: sliit-research-2026.	DONE
3	Data check (GEE)	15 low-cloud Sentinel-2 scenes (2025), Sentinel-1 radar on 26 Nov, 2 Dec, 8 Dec 2025, 349,477 Open Buildings footprints, SRTM DEM.	DONE
4	Cloud-free image	Cloud Score+ removed clouds pixel by pixel; 140 Sentinel-2 scenes combined into one clear pre-flood image.	DONE
5	Flood map	SAR change detection (before vs after), 4 rule variants compared; variant D (VV or VH) chosen; 12.47 km^2 flood water.	DONE
6	Labels	Exact distance from every building to flood water; 3 exposure classes: High ≤100 m, Medium 100–300 m, Low >300 m.	DONE
7	Features & patches	14 pre-flood features per building + 32×32-pixel (320 m) image patches with 10 bands.	DONE
8	Spatial folds	50,000 buildings, 144 blocks of 2 km × 2 km, grouped into 10 folds (no block is in train and test at the same time).	DONE
9	Models	10 models trained on the same folds: LR, RF, XGBoost, LightGBM, CNN from scratch, ResNet (random / ImageNet / SSL4EO), Dual-Stream, HSC-Attention.	DONE
10	Hypothesis tests	Friedman + Iman–Davenport, Nemenyi, Wilcoxon (Holm), McNemar, block-bootstrap CIs.	DONE
11	Demo	Local web app (index.html / run_demo.bat) with map of all buildings + comparison tables.	DONE
12	Model optimisation	Nested spatial-CV search (Optuna TPE) over HSC-Attention architecture (layers, heads, width, token grid, activation, attention direction, initialisation) and training settings; same budget for Baseline 1; ablations No-DEM, No-Cross-Attention, Visual-only (Section 9).	DONE
13	M4 integration	Replace "where flood happened" with M4's 3-day forecast flood probability as a model input.	NEXT
14	Second event	Validate on another flood (May 2016 Kelani / June 2024) to test transferability over time.	NEXT
15	Damage track	Train damage-grade model on xBD flood events (Florence, Harvey, Midwest) for the 4 damage classes.	NEXT
16	Full HSC-Attention	Repeat optimisation with the M4 forecast input; dashboard export (GeoJSON).	NEXT

2. The problem and the key ideas (theory in simple words)

2.1 Problem

During monsoon floods, Sri Lanka's Disaster Management Centre (DMC) needs to know which houses are at risk *before* or *while* water arrives. Field surveys take 2–4 weeks and optical satellites cannot see through monsoon cloud. M1 asks: **using only information available before a flood, can a model predict how exposed each building will be?**

2.2 Flood exposure vs. flood damage

Exposure = whether a building is in or near flood water. **Damage** = how badly the structure is broken. Exposure is the first and necessary condition for damage. Damage grades (No/Minor/Major/Destroyed) need sub-metre imagery and expert labelling; these do not exist for Sri Lanka. Therefore Progress 01 predicts exposure (measurable, Sri Lanka-specific), and the damage grade task is kept as a separate track using the xBD benchmark (Section 13).

2.3 Satellites used

- **Sentinel-2 (optical, 10 m):** a photograph-like image in visible and infrared bands. One house ≈ 1 pixel, so the model sees the *neighbourhood* (roofs, paddy, marsh, trees, river), not the house shape.
- **Sentinel-1 (radar/SAR, 10 m):** sends microwaves and measures the echo (backscatter, in dB). Calm water reflects the signal away, so it looks **dark**. Radar works through cloud and at night, which is why it gives the flood "answer".
- **Double-bounce:** in towns, the radar bounces water → wall → satellite, so flooded streets can look **bright**. This is why simple "dark = water" misses some urban flooding (a limitation we report).
- **SRTM DEM (30 m)** = ground height. **HAND** (MERIT Hydro) = Height Above the Nearest Drainage: how many metres a place is above the nearest stream. Low HAND = water reaches it first.
- **Google Open Buildings v3** = AI-drawn building footprints from 50 cm imagery (we kept confidence ≥ 0.75).

2.4 Study area and event

Lower Kelani basin box 79.85–80.22 °E, 6.90–7.02 °N (Colombo coast → Avissawella). Cyclone Ditwah formed over Sri Lanka on 27 Nov 2025 (Copernicus EMS activation EMSR851). The Kelani at Nagalagam Street reached the critical level of 8.45 ft on 1 Dec 2025 and fell to the minor flood level (6.9 ft) on the afternoon of 2 Dec 2025. Our flood image (Sentinel-1, 2 Dec, descending orbit) therefore shows a **receding** flood — an important assumption.

3. Data and label generation

3.1 Data inventory (verified in Earth Engine)

Data	Source / GEE ID	Use	Verified value
Sentinel-2 L2A	COPERNICUS/S2_SR_HARMONIZED + GOOGLE/CLOUD_SCORE_PLUS	Pre-flood image (Jan–Oct 2025)	140 scenes in composite; 15 scenes <20% cloud
Sentinel-1 GRD	COPERNICUS/S1_GRD (IW, VV+VH, descending)	Flood "answer" (2 Dec) vs reference (Oct–mid Nov)	Post dates 26 Nov, 2 Dec, 8 Dec 2025
Buildings	GOOGLE/Research/open-buildings/v3/polygons	One sample per building	349,477 (conf ≥ 0.75); 341,971 usable
Elevation	USGS/SRTMGL1_003	Height, slope	–11 to 367 m in box

Data	Source / GEE ID	Use	Verified value
HAND	MERIT/Hydro/v1_0_1 (hnd)	Height above drainage	—
Permanent water	JRC/GSW1_4/GlobalSurfaceWater	Remove rivers from flood map; distance to river	occurrence ≥ 50%
Land cover	ESA/WorldCover/v200	Feature	—

3.2 Flood map: SAR change detection and sensitivity test

A pixel is called **new flood water** when it became much darker than its normal (mean of Oct–mid-Nov 2025) and is now dark. Four rule variants were compared visually and numerically before choosing one (a *sensitivity analysis*):

Variant	Rule	Flood area (km^2)	Decision
A_current	dVV < –3 dB & VV < –16 dB & slope < 5°	6.87	Too strict; slope filter removes real water
B_noSlope	A without slope filter	8.35	—
C_relaxed	dVV < –2 dB & VV < –15 dB	9.88	—
D_VV_or_VH	C OR (dVH < –3 dB & VH < –22 dB)	10.80 (GEE) / 12.47 (export grid)	**Chosen**: best match to visible dark patches

All variants also remove permanent water (JRC occurrence ≥ 50 %) and patches smaller than 8 pixels. We deliberately did **not** use slope or HAND inside the final label rule, because those are also model inputs; using them in the label would let the model "learn our rule" (circular labelling).

3.3 From flood map to building labels

For every building we computed the exact Euclidean distance (in metres) from its centroid to the nearest flood pixel. The ordinal exposure classes are: **High** ≤ 100 m, **Medium** 100–300 m, **Low** > 300 m. ≤ 50 m was rejected as a class boundary because only 1.35 % of buildings fall in it (too few to learn).

Distance to flood water	≤ 0 m	≤ 30 m	≤ 50 m	≤ 100 m	≤ 150 m	≤ 200 m	≤ 300 m
% of 341,971 buildings	0.01	0.33	1.35	5.40	10.46	15.66	25.38

Physical sanity check (medians, near ≤ 50 m vs far > 150 m): HAND 4.40 vs 8.40 m; elevation 12 vs 19 m; distance to river 1,376 vs 1,841 m. Exposed buildings are lower and closer to drainage, which matches flood physics — evidence the labels are meaningful.

3.4 Assumptions (stated openly)

- A1. SAR flood water on 2 Dec 2025 represents flood exposure for Ditwah (it is ~1 day after the Kelani peak, so true exposure is likely **larger**).
- A2. Distance to flood water is a valid proxy for exposure; it is **not** a damage measurement.
- A3. Open Buildings footprints (conf ≥ 0.75) are correct enough; missing/false buildings add noise.
- A4. The 2025 cloud-free Sentinel-2 composite represents conditions just before the flood.
- A5. 2 km spatial blocks are large enough to reduce spatial leakage between training and test buildings.
- A6. The 50,000-building random sample represents the 341,971 buildings (same class proportions).
- A7. Thresholds (–2/–15 dB, –3/–22 dB, 100/300 m) are research choices, justified by the sensitivity test and class balance, not universal constants.

4. Models compared and why

We compare three families. All use exactly the same 50,000 buildings and the same 10 spatial folds, so differences come from the model, not the data.

ID	Model	Input	Pre-training	Why it is in the comparison
T1	Logistic Regression	14 tabular features	none	Simplest linear baseline
T2	Random Forest (300 trees)	14 tabular features	none	Strong classic model; best in Seleem et al. (2022)
T3	XGBoost	14 tabular features	none	Gradient boosting standard
T4	LightGBM	14 tabular features	none	Best in Sri Lankan study (Kurugama et al., 2024)
D1	Simple CNN (5 conv layers)	10-band 32×32 patch	**none (from scratch)**	Does a small CNN learn from images alone?
D2	ResNet-18	10-band patch	**none (random init)**	Same architecture as D3, isolates effect of pre-training
D3	ResNet-18	10-band patch	**ImageNet** (natural photos)	Classic transfer learning
D4	ResNet-50	RGB patch	**SSL4EO-S12** (Sentinel-2, MoCo)	Pre-trained on satellite images (Wang et al., 2023)
D5	Dual-Stream ResNet + concatenation	S2 stream + terrain stream	ImageNet (visual stream)	**Proposal Baseline 1**
D6	HSC-Attention (prototype)	S2 tokens attend to terrain tokens	ImageNet (visual stream)	**Proposed model** (first prototype)

4.1 How the main models work

- **Random Forest:** many decision trees, each trained on a random part of the data and random features; the forest votes. Averaging many noisy trees reduces variance (bagging).
- **XGBoost / LightGBM (boosting):** trees are added one by one; each new tree corrects the errors of the previous ones (gradient boosting). LightGBM grows trees leaf-wise and uses histograms, so it is fast.
- **CNN:** learns small filters (3×3) that detect patterns such as water edges, paddy texture or dense roofs, then combines them into higher-level features.
- **ResNet:** a deep CNN with "skip connections" (output = F(x) + x) that make very deep networks trainable.
- **Transfer learning / fine-tuning:** start from weights already trained on a large dataset (ImageNet photos, or SSL4EO Sentinel-2 images) and continue training on our task. Fine-tuning means *all* weights are updated with a small learning rate (3×10^{-4}). For 10-band inputs, the first layer was widened: RGB filters were copied to B4/B3/B2 and the mean filter was used for other bands.
- **Dual-Stream + concatenation (D5):** stream A (ResNet-18) encodes Sentinel-2; stream B (4-layer CNN) encodes terrain (DEM, slope, HAND, river distance, building mask). The two feature vectors are simply joined (concatenated) and passed to a classifier.
- **HSC-Attention (D6):** instead of joining, each of the 16 visual tokens (4×4 grid, 256-dim) asks the 16 hydro-terrain tokens "which places matter for me?" using cross-attention: Attention(Q,K,V) = $\text{softmax}(QK^T/\sqrt{d})V$, with Q from the image and K,V from terrain, 4 heads, followed by residual + LayerNorm + feed-forward. This lets the model weight terrain differently for different parts of the neighbourhood.

What is technically different between D5 and D6? Only the fusion step: D5 = global-average then concatenate (no spatial interaction); D6 = token-level cross-attention (spatially-aware interaction). So D6 vs D5 is exactly the proposal's "No-Cross-Attention" ablation.

5. Experimental design

5.1 Spatial cross-validation (why not random split?)

Neighbouring buildings look alike and have similar labels (spatial autocorrelation). If neighbours are split between training and test, the model is tested on "almost seen" data and the score is too optimistic (Ploton et al., 2020). We therefore cut the area into 2 km × 2 km blocks (144 blocks) and assigned **whole blocks** to 10 folds. Each fold is used once as the test set. We also ran a random-split CV for the best tabular model to measure how optimistic random splits would be (Section 7.3).

5.2 Why 10 folds and not 5?

The Wilcoxon signed-rank test with n = 5 paired folds can never give $p < 0.05$ (the smallest possible two-sided p is 0.0625). With n = 10 the smallest one-sided p is $1/1024 \approx 0.001$, so real differences can be detected. This is why the proposal's 5-fold plan was changed to 10 folds.

5.3 Training settings (fixed, same for all deep models)

- Sample: 50,000 buildings (seed 42); class counts Low 37,462 · Medium 9,844 · High 2,694.
- Class imbalance handled with inverse-frequency class weights in the loss (all models).
- Deep models: 6 epochs, batch 256, AdamW (weight decay 1e-4), one-cycle learning-rate schedule, mixed precision, flips + 90° rotations augmentation; patches upsampled 32→64 px for ResNets; per-fold band normalisation computed from training blocks only (no leakage).
- Tabular: 14 features (S2 bands B2,B3,B4,B8,B11, NDVI, NDWI, DEM, slope, HAND, distance to river, building density within 100 m, land cover, footprint area).
- No hyper-parameter tuning on the test folds. Fixed reasonable defaults were used for every model so that the comparison is fair — Section 9 then optimises D6 and Baseline 1 with nested spatial CV.

5.4 Metrics

Metric	Meaning (simple)	Why
Macro-F1 (primary)	Average of F1 for Low, Medium, High (each class counts equally)	Fair under imbalance; proposal metric
Balanced accuracy	Average recall of the three classes	Shows how many of each class are found
QWK	Quadratic-weighted kappa: agreement beyond chance, big mistakes (Low↔High) penalised more	Classes are ordered (ordinal)
AUC (High vs rest)	Probability the model ranks a High building above a non-High one	Comparable to flood-susceptibility papers
Latency	Milliseconds to predict 1,000 buildings	Operational speed (NFR-2.1)

IoU and MAE from the proposal are not used: IoU needs a segmentation output and MAE needs true damage percentages — neither exists for this task.

6. Hypothesis testing — the theory used to select the best model

A higher average score can happen by chance. Hypothesis testing tells us whether the difference is larger than random fold-to-fold variation. We follow the method recommended by Demšar (2006) for comparing several models over several test sets.

Test	Null hypothesis H0	What it uses	Decision rule
Friedman	All models have equal performance (equal mean ranks)	Rank of each model in each of the 10 folds	$p < 0.05 \rightarrow$ at least one model differs
Iman–Davenport	Same as Friedman (less conservative F version)	Friedman χ^2 converted to F	$p < 0.05 \rightarrow$ reject H0
Nemenyi post-hoc	Two models have equal mean rank	Critical difference CD = $q_\alpha \sqrt{(k(k+1)/6n)}$	Rank gap > CD → different
Wilcoxon signed-rank (Holm)	Best model is not better than model j	10 paired fold differences in macro-F1	Holm-adjusted $p < 0.05 \rightarrow$ best is better
McNemar	Best and runner-up make the same number of errors	Buildings where only one of the two is correct (b vs c)	$p < 0.05 \rightarrow$ error rates differ
Block bootstrap 95% CI	—	1,000 resamples of whole spatial blocks	Non-overlapping CIs = clear separation

- **p-value:** the probability of seeing a difference at least this large if H0 were true. Small p = evidence against H0.
- α **= 0.05:** we accept a 5 % risk of a false "better" claim. With many pairwise tests we use the **Holm correction** to keep this risk at 5 % overall.
- **Why non-parametric tests:** fold scores are few (10) and not guaranteed normal; rank-based tests do not assume normality.
- **Selection rule used:** the selected model is the one with the best mean rank / macro-F1; we then report whether it is *statistically* better than each other model. If not significant, we say the models are statistically tied and prefer the simpler/faster one.

7. Results

7.1 Main comparison table (spatial 10-fold CV, measured)

Model	Macro-F1 (mean ± SD)	Bal. acc.	QWK	AUC High	Mean rank	Train s / fold	ms / 1000 bldgs
D6_HSC_Attention	0.666 ± 0.030	0.745	0.594	0.963	1.50	51.1	70.4
D2_ResNet18_random	0.648 ± 0.039	0.744	0.572	0.959	3.40	44.3	143.1
D1_SimpleCNN_scratch	0.644 ± 0.041	0.734	0.575	0.953	3.40	17.6	48.3
D3_ResNet18_ImageNet	0.642 ± 0.037	0.724	0.559	0.958	3.20	43.2	55.3
D5_DualStream_Concat	0.636 ± 0.044	0.719	0.556	0.957	3.50	56.8	185.2
T2_RandomForest	0.551 ± 0.034	0.569	0.439	0.880	6.60	37.9	49.8
T4_LightGBM	0.543 ± 0.029	0.592	0.419	0.878	7.10	7.7	166.5
T3_XGBoost	0.531 ± 0.030	0.615	0.398	0.881	8.30	7.6	23.0
D4_ResNet50_SSL4EO	0.528 ± 0.057	0.639	0.353	0.928	8.10	107.5	299.7
T1_LogReg	0.473 ± 0.039	0.603	0.304	0.870	9.90	2.7	1.3

Highlighted row = selected model. Latency for tabular models is CPU time in Colab (2 vCPU); for deep models it is T4 GPU time. Training time is per fold (about 45,000 training buildings).

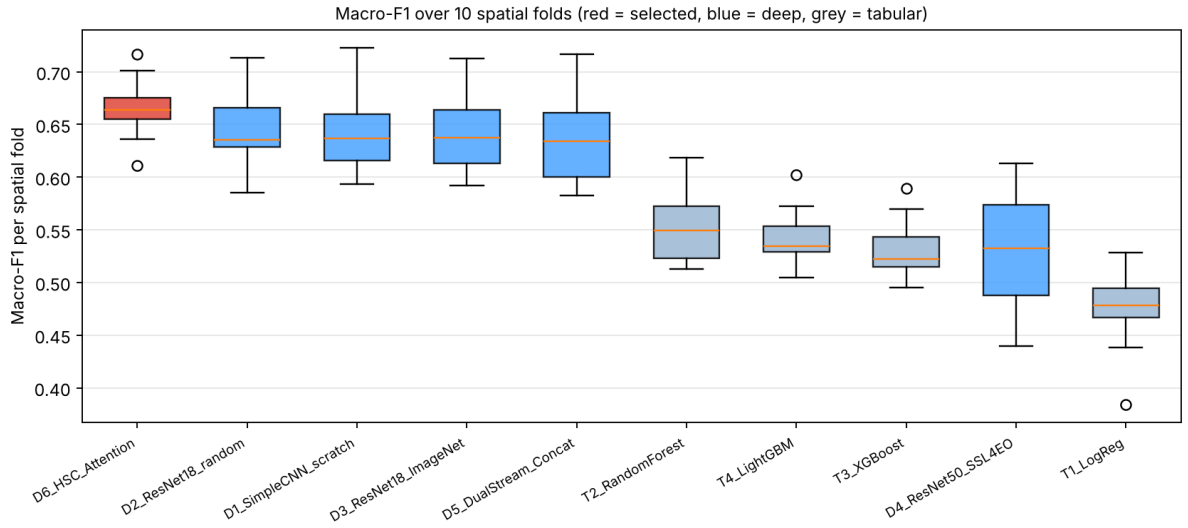


Figure 1. Macro-F1 of every model in each of the 10 spatial folds. Each box = 10 numbers.

7.2 Does pre-training help? (same architecture, different starting weights)

ResNet-18 trained from random weights (D2) reached macro-F1 0.648, while the same ResNet-18 starting from ImageNet weights (D3) reached 0.642; the satellite-pretrained ResNet-50 (SSL4EO-S12, D4, RGB only) reached 0.528. The small CNN trained from scratch (D1) reached 0.644. So pre-training did not give a clear gain here: with 45,000 labelled buildings per fold and 10-band 32x32 inputs, the network can learn useful filters from scratch, and ImageNet filters (learned on natural RGB photos) transfer only partly to multispectral 10 m imagery. D4 uses only 3 RGB bands (the available SSL4EO weights are RGB), so it cannot use NIR/SWIR or terrain bands; its result should be read with this in mind.

7.3 Random split vs spatial split (why spatial CV matters)

For T2_RandomForest, a random 10-fold split gives macro-F1 0.605 and AUC(High) 0.910, while the spatial split gives 0.551 and 0.880. The difference (+0.054 macro-F1) is the optimism that a random split would have added. This is why our numbers should not be compared one-to-one with papers that used random splits.

7.4 What the model learned (feature importance, LightGBM gain)

Feature	Share of total gain
X_built_frac_100m	0.171
X_dem_m	0.169
X_dist_river_m	0.156
X_hand_m	0.103
X_slope_deg	0.077
X_B11	0.049
X_area_m2	0.046
X_B8	0.044

The most important features are X_built_frac_100m (17%), X_dem_m (17%), X_dist_river_m (16%), X_hand_m (10%), X_slope_deg (8%). Terrain-related features (HAND, elevation, distance to river) and land-surface wetness signals matter most, which agrees with flood physics.

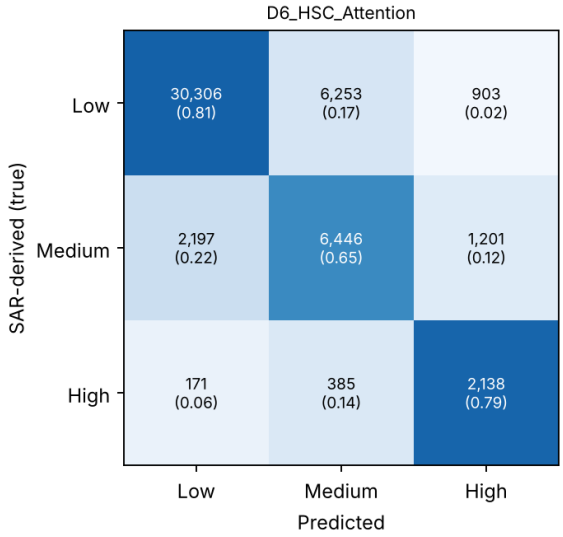


Figure 2. Pooled out-of-fold confusion matrix of D6_HSC_Attention (rows = SAR-derived class, columns = prediction).

8. Hypothesis-testing results and model selection

Friedman test: χ^2 = 78.37, p = 3.40e-13; Iman–Davenport F = 60.65, p = 1.11e-16 (k = 10 models, n = 10 folds). **H0 is rejected**: the models do not all perform the same.

Nemenyi critical difference (α = 0.05): CD = 4.28 rank units. Models whose mean ranks differ by more than this are significantly different.

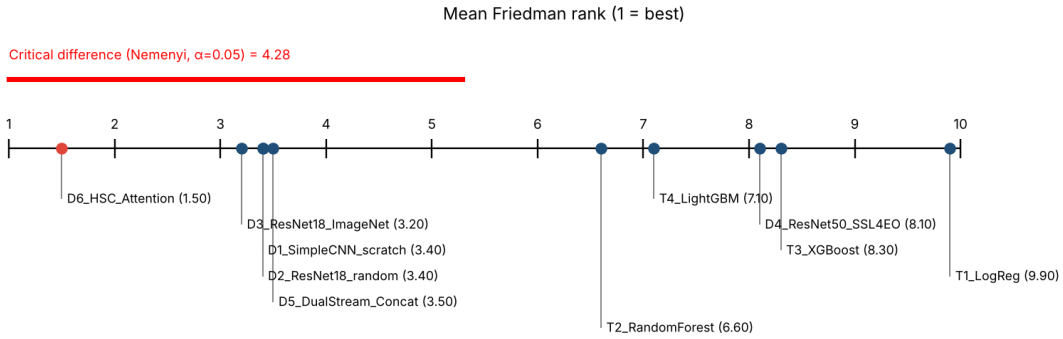


Figure 3. Critical-difference diagram (mean Friedman rank; lower is better; red bar = CD).

Pairwise Wilcoxon signed-rank tests (one-sided H1: selected model is better), Holm-corrected:

Selected model vs	Mean diff. F1	Folds won	p (raw)	p (Holm)	Significant at 0.05?
D4_ResNet50_SSL4EO	+0.138	10/10	9.77e-04	0.0088	Yes
T3_XGBoost	+0.134	10/10	9.77e-04	0.0088	Yes
T2_RandomForest	+0.114	10/10	9.77e-04	0.0088	Yes
T1_LogReg	+0.193	10/10	9.77e-04	0.0088	Yes
T4_LightGBM	+0.123	10/10	9.77e-04	0.0088	Yes
D3_ResNet18_ImageNet	+0.023	9/10	0.0029	0.0117	Yes
D5_DualStream_Concat	+0.029	9/10	0.0049	0.0146	Yes
D1_SimpleCNN_scratch	+0.021	8/10	0.0068	0.0146	Yes
D2_ResNet18_random	+0.018	9/10	0.0068	0.0146	Yes

McNemar test (D6_HSC_Attention vs runner-up D2_ResNet18_random) on all 50,000 out-of-fold predictions: b = 3904 buildings only D6_HSC_Attention got right, c = 3158 only D2_ResNet18_random got right; χ^2 = 78.59, p < 1e-16.

Model	Pooled macro-F1	95% CI (block bootstrap)
D6_HSC_Attention	0.681	0.648 – 0.708
D2_ResNet18_random	0.661	0.626 – 0.691
D1_SimpleCNN_scratch	0.656	0.623 – 0.686
D3_ResNet18_ImageNet	0.661	0.628 – 0.691
D5_DualStream_Concat	0.657	0.627 – 0.687
T2_RandomForest	0.569	0.542 – 0.598
T4_LightGBM	0.559	0.530 – 0.586
T3_XGBoost	0.547	0.513 – 0.576
D4_ResNet50_SSL4EO	0.542	0.505 – 0.573
T1_LogReg	0.490	0.456 – 0.522

Decision. D6_HSC_Attention has the highest mean macro-F1 (0.666) and the best mean rank (1.50). Friedman rejects equal performance (p = 3.40e-13); Wilcoxon tests show that the selected model is significantly better than 9 of 9 other models after Holm correction. McNemar vs D2_ResNet18_random: p < 1e-16. Best tabular model: T2_RandomForest (0.551); best deep model: D6_HSC_Attention (0.666). Therefore D6_HSC_Attention is selected as the Progress-01 model. **Caution:** the all-pairs Nemenyi test is more conservative: the rank gap between D6_HSC_Attention and D2_ResNet18_random, D1_SimpleCNN_scratch, D3_ResNet18_ImageNet, D5_DualStream_Concat is smaller than the critical difference (4.28), so Nemenyi cannot separate D6_HSC_Attention from these models; only the planned best-vs-others Wilcoxon tests (Holm-corrected) do. We therefore say: HSC-Attention is the best model in this experiment, with a small but consistent margin (it won 8–9 of 10 folds against each deep model), and it is clearly better than all tabular models. The proposed HSC-Attention prototype (D6, 0.666) vs the proposal Baseline 1 Dual-Stream concatenation (D5, 0.636): difference +0.029. Cross-attention helped slightly in this first prototype; without the M4 forecast input its main advantage (spatially weighting forecast flood depth) cannot yet be shown.

9. Model optimisation of HSC-Attention (algorithmic contribution)

9.1 What is optimised and why

A neural network has two kinds of numbers. **Parameters** (weights) are learned by gradient descent. **Hyper-parameters** are chosen by the researcher before training: the *architecture* (how many cross-attention layers, how many attention heads, token width, token grid, activation function, one-way or two-way attention) and the *training recipe* (learning rate, weight decay, dropout, epochs, starting weights). In Step 8 these were fixed by hand. Here they are searched systematically, which is the algorithmic-optimisation contribution of M1.

9.2 Theory: why nested spatial cross-validation

- If settings are chosen by looking at the test folds, the reported score is optimistically biased (selection bias; Cawley & Talbot, 2010). Nested cross-validation (Varma & Simon, 2006) removes this: an **inner** loop chooses settings, an **outer** loop measures performance on data never used for choosing.
- Outer loop: the same 10 spatial folds as Steps 7–8. For outer fold k, the inner validation set is fold k+1 (whole 2 km blocks) and the inner training set is a 50 % random subsample of the other 8 folds (a common low-fidelity speed-up). Test fold k is never seen during the search.
- Search algorithm: Optuna (Akiba et al., 2019) with the Tree-structured Parzen Estimator (TPE; Bergstra et al., 2011): 14 trials planned per outer fold for D6 and 8 for D5 (effective real trials: see the protocol deviation in 9.8). The first 5 trials are random (random search is a strong baseline; Bergstra & Bengio, 2012). Selection criterion: macro-F1 on the inner validation fold.
- After the search, the best setting is re-trained on all 9 outer-training folds and tested once on fold k. This gives 10 unbiased outer scores per model, paired with the defaults (re-run with the same code and seeds) for Wilcoxon tests.
- **Fairness:** Baseline 1 (D5 Dual-Stream) receives the same procedure. Optimising only the proposed model would make any comparison unfair.

9.3 Search space

Setting	Values searched	Default (Step 8)	Chosen in the 10 outer folds (count)
learning rate	1e-4 – 2e-3 (log)	3e-4	median 0.00056 (range 0.00017 – 0.0014)
weight decay	1e-5 – 1e-2 (log)	1e-4	median 9e-05 (range 1.9e-05 – 0.0052)
epochs	6, 10, 14	6	14: 6, 10: 2, 6: 2
classifier dropout	0.1 – 0.5	0.3	median 0.26 (range 0.1 – 0.49)
visual-stream initialisation	ImageNet, random	imagenet	imagenet: 10
token width d	128, 256	256	128: 5, 256: 5
attention heads	2, 4, 8	4	2: 4, 4: 4, 8: 2
cross-attention layers	1, 2, 3	1	1: 4, 3: 4, 2: 2
FFN activation	GELU, ReLU, SiLU	gelu	relu: 6, silu: 3, gelu: 1
token grid	4×4, 8×8 tokens	4	4: 7, 8: 3
bidirectional attention	no, yes	False	True: 7, False: 3

Activation functions: ReLU = max(0, x); GELU = x·Φ(x), a smooth version used in Transformers (Hendrycks & Gimpel, 2016); SiLU = x·sigmoid(x) (Elfwing et al., 2018). Token grid 8×8 uses ResNet stage 2 (finer, 64 tokens of 40 m) instead of stage 3 (16 tokens of 80 m). Bidirectional = terrain tokens also attend back to the image tokens in every layer.

Baseline 1 (D5) search space and choices:

Setting	Values searched	Default (Step 8)	Chosen in the 10 outer folds (count)
learning rate	1e-4 – 2e-3 (log)	3e-4	median 0.00085 (range 0.00029 – 0.0019)
weight decay	1e-5 – 1e-2 (log)	1e-4	median 8.2e-05 (range 1.1e-05 – 0.0047)
epochs	6, 10, 14	6	10: 6, 6: 2, 14: 2
classifier dropout	0.1 – 0.5	0.3	median 0.24 (range 0.11 – 0.48)
visual-stream initialisation	ImageNet, random	imagenet	imagenet: 8, random: 2
classifier hidden units	128, 256, 512	256	512: 4, 128: 3, 256: 3
classifier activation	ReLU, GELU, SiLU	relu	gelu: 5, silu: 3, relu: 2

9.4 Results (outer spatial 10-fold CV, measured)

Model	Macro-F1 (mean ± SD)	QWK	AUC High	Params (M)	Train s / fold
D6 HSC-Attention (optimised)	0.676 ± 0.037	0.604	0.956	4.11	125
D6 HSC-Attention (default)	0.660 ± 0.035	0.584	0.960	4.19	53
D5 Dual-Stream (optimised)	0.650 ± 0.039	0.573	0.956	12.17	98
D5 Dual-Stream (default)	0.645 ± 0.038	0.569	0.959	12.12	58
A1 No-DEM (no elevation, slope, HAND)	0.604 ± 0.038	0.482	0.939	4.10	121
A2 No-Cross-Attention (concat)	0.661 ± 0.038	0.585	0.955	2.81	92
A3 Visual-only (no terrain stream)	0.563 ± 0.058	0.415	0.934	2.23	63

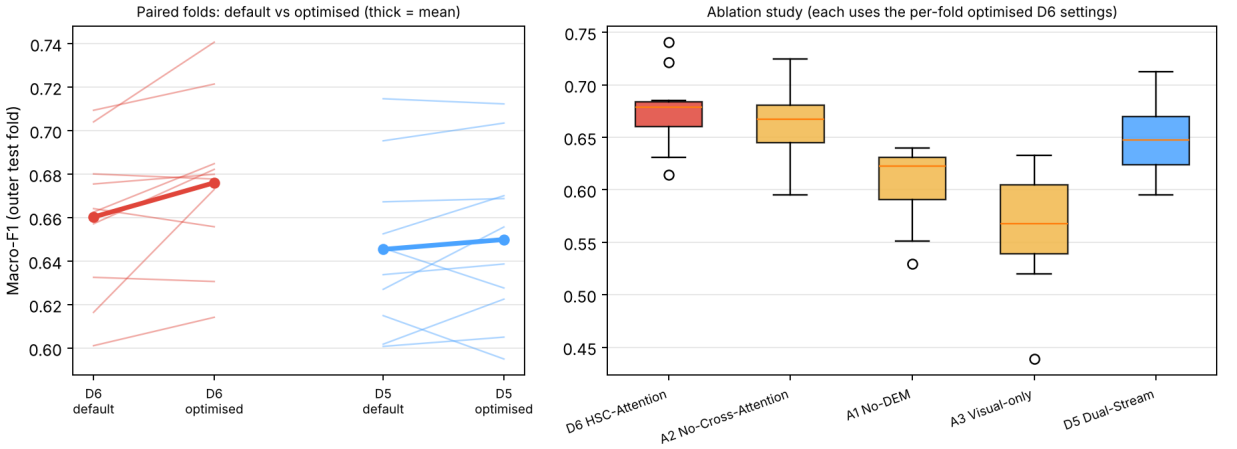


Figure 4. Left: each thin line is one outer test fold (default → optimised). Right: ablations of the optimised HSC-Attention.

9.5 Hypothesis tests (two-sided Wilcoxon signed-rank, Holm correction over the 6 planned comparisons)

Planned comparison	Mean diff. F1	Folds won	p (raw)	p (Holm)	Significant?
D6 HSC-Attention (opt.) vs D6 HSC-Attention (default)	+0.016	7/10	0.0371	0.0742	No
D5 Dual-Stream (opt.) vs D5 Dual-Stream (default)	+0.005	7/10	0.3223	0.3223	No
D6 HSC-Attention (opt.) vs D5 Dual-Stream (optimised)	+0.026	10/10	0.0020	0.0117	Yes
D6 HSC-Attention (opt.) vs A1 No-DEM (no elevation, slope, HAND)	+0.072	10/10	0.0020	0.0117	Yes

Planned comparison	Mean diff. F1	Folds won	p (raw)	p (Holm)	Significant?
D6 HSC-Attention (opt.) vs A2 No-Cross-Attention (concat)	+0.015	8/10	0.0098	0.0293	Yes
D6 HSC-Attention (opt.) vs A3 Visual-only (no terrain stream)	+0.113	10/10	0.0020	0.0117	Yes

Interpretation. Optimisation changed HSC-Attention from macro-F1 0.660 (default) to 0.676 (optimised), a difference of +0.016; the optimised version won 7/10 outer folds (Wilcoxon, Holm-adjusted p = 0.0742). This change is **not** statistically significant, so we cannot claim that tuning improved the model; the default settings were already close to the best found within the budget. Baseline 1 changed from 0.645 to 0.650 (+0.005; Holm p = 0.3223). After both models received the **same** optimisation budget, HSC-Attention (0.676) vs Dual-Stream (0.650): difference +0.026, won 10/10 folds, Holm p = 0.0117 — the cross-attention model is significantly better under a fair comparison.

Friedman test over all 12 models (Step 7–8 models plus the two optimised models): χ^2 = 95.08, p = 1.68e-15; Nemenyi CD = 5.27. Best mean ranks: D6_tuned 1.50; D6_HSC_Attention 2.50; D5_tuned 4.30; D1_SimpleCNN_scratch 4.80; D3_ResNet18_ImageNet 4.80.

Model	Pooled out-of-fold macro-F1	95% CI (bootstrap over spatial folds)
D6 HSC-Attention (optimised)	0.692	0.661 – 0.719
D6 HSC-Attention (default)	0.678	0.650 – 0.702
D5 Dual-Stream (optimised)	0.667	0.631 – 0.697

9.6 Ablation study (does each part of HSC-Attention matter?)

- **A1 No-DEM (no elevation, slope, HAND):** removing elevation, slope and HAND significantly reduced macro-F1 (full model minus ablation = +0.072, full model won 10/10 folds, Holm p = 0.0117).
- **A2 No-Cross-Attention (concat):** replacing cross-attention by concatenation significantly reduced macro-F1 (full model minus ablation = +0.015, full model won 8/10 folds, Holm p = 0.0293).
- **A3 Visual-only (no terrain stream):** removing the whole terrain stream significantly reduced macro-F1 (full model minus ablation = +0.113, full model won 10/10 folds, Holm p = 0.0117).

Each ablation keeps the optimised settings of its outer fold and removes exactly one component, so any change is caused by that component.

9.7 Which settings mattered?

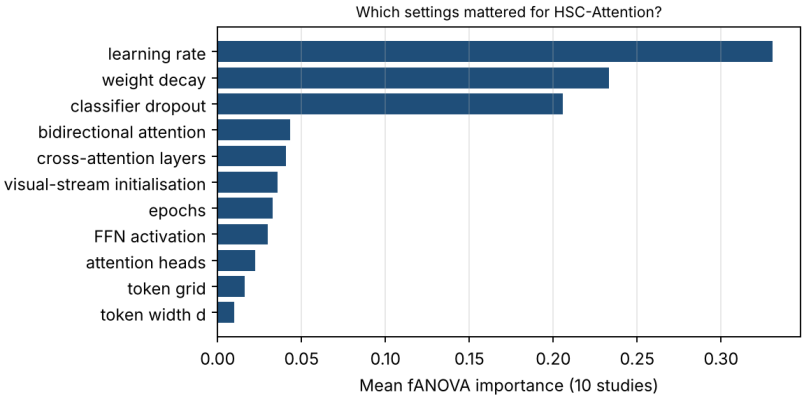


Figure 5. Hyper-parameter importance (fANOVA; Hutter et al., 2014), averaged over the 10 inner studies. With only a few trials per study these values are indicative, not precise.

9.8 Honest reading of the optimisation result

- **Protocol deviation (found by our own audit and reported openly).** A synthetic-data smoke test, run just before the real search in the same Colab runtime, left temporary Optuna study files. The first real session re-used them, so most inner studies contain 2 synthetic-data trials. Effective real trials per outer fold: D6 12 (planned 14), D5 6/8 (planned 8). Consequences: (i) the search budget was slightly smaller than planned and the default setting was not re-evaluated inside most inner searches; (ii) the parameter-importance values above use real trials only; (iii) **outer test results are not affected** — the synthetic trials never saw the test folds, every chosen setting came from a real trial (verified for all 20 studies), and the defaults were evaluated separately on the same outer folds. The final-model search ran in a fresh runtime and is clean (14 real trials).
- Reproducibility check: the default HSC-Attention re-run with the Step 12 code scored 0.660 vs 0.666 in Step 8 (same data, folds, seeds; small differences come from GPU non-determinism).
- **Final deployed setting** (chosen by the same procedure on the whole sample): 2 cross-attention layer(s), 8 heads, d = 256, 4×4 token grid, GELU FFN, one-way (image → terrain) attention, imagenet initialisation, learning rate 5.1e-04, weight decay 1.1e-05, dropout 0.32, 14 epochs.

10. Comparison with published research

Study	Task / data	Validation	Best model	Reported result
Kurugama et al. (2024), J. Flood Risk Management, doi:10.1111/jfr3.12980	Flood susceptibility, Rathnapura (Sri Lanka); 595 inventory points, 13 factors	Train/test split (not spatially blocked)	LightGBM (vs GBM, XGBoost, CatBoost, LogitBoost)	AUC 0.970, F1 0.907, κ 0.813
Seleem et al. (2022), Geomatics, Natural Hazards & Risk 13(1)	Urban pluvial flood susceptibility, Berlin; CNN, ANN, RF, SVM; 2–30 m	Test set + transfer to another area	Random Forest; CNN transferred best	All AUC ≥ 0.87; transfer AUC ≥ 0.80
Gupta et al. (2019), CVPR-W (xBD)	Building damage (4 classes) from pre+post 0.5 m imagery	Official split	Baseline CNN	Different task (post-event damage)
This work (M1)	Building-level flood exposure, lower Kelani; 341,971 buildings; pre-flood inputs only	Spatial 10-fold block CV + hypothesis tests	D6_HSC_Attention / D6 optimised (Sec. 9)	Macro-F1 0.666; AUC(High) 0.963 · optimised: 0.676; 0.956

How to read this comparison. The Sri Lankan susceptibility study reports AUC 0.970 with a non-spatial split; our best AUC(High) under spatial CV is 0.963. Our own experiment shows that a random split inflates scores (for T2_RandomForest: AUC 0.910 random vs 0.880 spatial; macro-F1 0.605 vs 0.551). Also, published studies classify flood/non-flood points (2 classes), while we predict 3 ordinal classes for every building, which is harder. Our AUC is in the same range as Seleem et al. (≥ 0.87), but the tasks differ, so this is context, not a competition.

The proposal's targets (macro-F1 ≥ 0.78, MAE ≤ 12 %, IoU ≥ 0.75) were set for a damage-grade task. They are **not met and not directly applicable** to the exposure task; we report this honestly. The claim "Baseline 1 ResNet F1 of 0.69" in the proposal could not be traced to a source and should be removed.

11. Final model and local web demo

The selected model (D6_HSC_Attention) was re-trained on all 50,000 sample buildings and then applied to all 341,971 buildings in 71.8 s (209.8 ms per 1,000 buildings). On the 291,971 buildings that were never in the 50,000-building sample, agreement with the SAR-derived class is macro-F1 0.758 (note: these buildings share neighbourhoods with training buildings, so this is not a spatial test). Predicted classes (Low / Medium / High): 226,645 / 85,489 / 29,837; SAR-derived: 255,169 / 68,335 / 18,467.

Updated after optimisation (Step 12): the demo now uses the optimised model D6_tuned with settings chosen by the same nested procedure on the whole sample, re-trained on all 50,000 buildings. It predicted 341,971 buildings in 78.7 s (230.3 ms per 1,000); macro-F1 on non-sample buildings 0.813. Predicted Low / Medium / High: 234,096 / 81,721 / 26,154.

The folder **M1_demo** contains **index.html**, **run_demo.bat** and an **assets** folder. It works offline.

- Option A: double-click **index.html** (Chrome or Edge).
- Option B (local web server): double-click **run_demo.bat** → it runs `python -m http.server 8000` and opens http://localhost:8000.
- The map shows all 341,971 buildings on the cloud-free Sentinel-2 image. Colour by model prediction, SAR label, correct/wrong, or P(High). Toggle the SAR flood-water layer. Click a building to see its prediction, SAR label, distance to flood and HAND.
- Below the map: the model comparison table, bar chart, hypothesis-test results, figures, literature comparison and limitations.

Suggested 3-minute demo script: (1) show the Sentinel-2 image and SAR flood layer; (2) switch to "SAR label" then "Model prediction" — point at Kolonnawa/Kaduwela where High exposure clusters; (3) switch to

"Correct / wrong" and explain errors honestly (urban double-bounce, receding flood); (4) scroll to the comparison table and Friedman/Wilcoxon results; (5) end with limitations and next steps.

12. Limitations and threats to validity

- **Label proxy:** exposure (distance to SAR water), not damage. No damage percentages exist, so MAE is not measurable.
- **Timing:** the 2 Dec SAR image is ~1 day after the Kelani peak; true exposure was larger, so some exposed buildings are labelled Low (label noise, biased towards under-labelling High).
- **Urban SAR blindness:** double-bounce hides water between dense buildings (1.47 % of buildings showed strong VV brightening).
- **One event, one basin:** results may not transfer to other floods or rivers without re-validation.
- **Resolution:** Sentinel-2 10 m cannot see individual roofs; SRTM 30 m DEM has metre-level vertical errors in towns.
- **No M4 forecast yet:** the forecast flood probability, the key input of the proposed HSC-Attention design, is not yet available; the D6 prototype uses terrain only.
- **Optimisation budget:** only 14 (D6) / 8 (D5) trials per outer fold on a 50 % inner subsample (free GPU); a larger budget might find better settings. Tabular models and D1–D4 were not tuned.
- **Sample:** models were compared on 50,000 of 341,971 buildings; the final model predicts all buildings.

13. Next steps after Progress 01

Priority	Task	Expected output
1	Get M4 3-day flood probability raster for Ditwah (or simulate with HAND-based inundation for prototype)	Hydro stream with forecast input → true "predictive" HSC-Attention
2	Second event (May 2016 Kelani / June 2024)	Temporal transfer test: train Ditwah, test other flood
3	Larger optimisation budget (more trials, full inner data) and tuning of the best tabular model	Tighter, fully fair comparison
4	Repeat the ablations once the M4 forecast input is added (No-Forecast ablation)	Evidence that forecast fusion helps
5	xBD flood events damage-grade track	4-class damage model (supports the proposal's damage objective)
6	Validate labels with Copernicus EMS EMSR851 / UNOSAT delineations	Independent label quality check
7	GeoJSON export + dashboard integration with M2–M4	System integration (FR-1.6, FR-1.7)

14. Likely panel questions and answers

Q: Why did you change from damage grades to flood exposure?

A: Because no building-damage labels exist for Sri Lanka and Sentinel-2 (10 m) cannot see roof damage. Exposure is measurable from Sentinel-1 radar and is the first condition for damage. The damage-grade task continues as a separate track using xBD.

Q: How do you know your labels are correct?

A: We compared four SAR rules (sensitivity test) and chose the one that best matched the visible flood; the labels agree with physics (exposed buildings have median HAND 4.4 m vs 8.4 m). Limitations: receding flood on 2 Dec and urban double-bounce. Next step: check against Copernicus EMS EMSR851 delineations.

Q: Is there data leakage?

A: We used spatial block CV (2 km blocks), per-fold normalisation from training blocks only, and kept terrain variables out of the label rule. All inputs are pre-flood. We also measured how much a random split would inflate scores (0.605 vs 0.551 macro-F1).

Q: Why macro-F1 and not accuracy?

A: 75% of buildings are Low exposure; a model that always says Low gets 75% accuracy but is useless. Macro-F1 gives each class equal weight, so High exposure counts.

Q: Why the Friedman test and not a t-test?

A: We compare many models over the same 10 folds. Friedman is the recommended non-parametric test for this (Demšar, 2006); it does not assume normal scores. Pairwise Wilcoxon tests with Holm correction then tell which differences are real.

Q: Why 10 folds instead of the 5 in your proposal?

A: With 5 folds the Wilcoxon test can never reach $p < 0.05$; 10 folds give enough paired results for a valid test.

Q: Did pre-training help?

A: ResNet-18 trained from random weights (D2) reached macro-F1 0.648, while the same ResNet-18 starting from ImageNet weights (D3) reached 0.642; the satellite-pretrained ResNet-50 (SSL4EO-S12, D4, RGB only) reached 0.528. The small CNN trained from scratch (D1) reached 0.644. So pre-training did not give a clear gain here: with 45,000 labelled buildings per fold and 10-band 32x32 inputs, the network can learn useful filters from scratch, and ImageNet filters (learned on natural RGB photos) transfer only partly to multispectral 10 m imagery. D4 uses only 3 RGB bands (the available SSL4EO weights are RGB), so it cannot use NIR/SWIR or terrain bands; its result should be read with this in mind.

Q: What is new in HSC-Attention compared with the baseline?

A: Baseline 1 joins image and terrain features after global pooling (no spatial interaction). HSC-Attention lets each image region attend to terrain regions with cross-attention, so terrain can be weighted differently across the neighbourhood. In this prototype: D6 0.666 vs D5 0.636.

Q: Your proposal targeted macro-F1 ≥ 0.78. Why is your result lower?

A: The target was for damage grades on xBD-style data. Our task is 3-class exposure from 10 m imagery with noisy SAR labels, tested spatially. The measured value (0.666) is honest; we will improve it with M4 forecast input, better labels and tuning.

Q: How fast is the model?

A: The final model predicted all 341,971 buildings in 71.8 s (209.8 ms per 1,000 buildings), far below the 45 s per tile requirement.

Q: Can DMC use this tomorrow?

A: Not yet. It is validated on one event and one basin. It needs a second event test and M4 forecast input. Today it is a research prototype with a working demo.

Q: What is your algorithmic contribution, and is the improvement real?

A: The HSC-Attention architecture (cross-attention depth, heads, width, token grid, activation, attention direction, initialisation) and its training settings were optimised with Optuna TPE inside nested spatial cross-validation, so test folds were never used for choosing. Baseline 1 got the same budget. Optimisation changed HSC-Attention from macro-F1 0.660 (default) to 0.676 (optimised), a difference of +0.016; the optimised version won 7/10 outer folds (Wilcoxon, Holm-adjusted p = 0.0742). This change is not statistically significant, so we cannot claim that tuning improved the model; the default settings were already close to the best found within the budget. Baseline 1 changed from 0.645 to 0.650 (+0.005; Holm p = 0.3223). After both models received the same optimisation budget, HSC-Attention (0.676) vs Dual-Stream (0.650): difference +0.026, won 10/10 folds, Holm p = 0.0117 — the cross-attention model is significantly better under a fair comparison.

Q: Why not just tune on the test folds, it is faster?

A: Because the reported score would then be optimistically biased (selection bias). Our own numbers show it: the inner-validation gain was larger than the outer-test gain.

15. Reproducibility record

Item	Value
Notebook	Google Drive: M1_Research_Main.ipynb (steps 6–10); Untitled21.ipynb (steps 2–5, GEE exploration)
GEE project	sliit-research-2026
GEE export task IDs	Raster stack 3QTMHS34SYPVPNRMMOU36N3B · Label raster W2FKP3R7YTK2VFCLBNQC4E24 · Building points GT56UEQQHY3FCCRN2RE4HGR5
Data files (Drive/M1_Research)	M1_raster_stack_v1.tif, M1_label_raster_v2.tif, M1_building_points_v2.csv, M1_buildings_dataset_v2.parquet, M1_sample_with_folds.parquet
Results (Drive/M1_Research/results)	cv_tabular.csv, cv_deep.csv, oof_*.npz, hypothesis_tests.json, model_summary.csv, nemenyi_pvalues.csv, figures/, predictions_all_buildings.parquet
Step 12 optimisation	step12_optimise.py; results/tuning/: outer_results.csv, best_params.json, trials_*.csv, D*_outer*.db (Optuna), optimisation_summary.json, final_config.json, final_model_record.json; results/final_model_optimised.pt; step12_log.txt
Random seed	42 (sample, folds, models); fold k uses seed 42+k for deep models
Software	Python 3.13 (Colab), PyTorch 2.x, torchvision, timm, torchgeo, scikit-learn 1.6.1, XGBoost 3.4.1, LightGBM 4.6.0, scikit-posthocs, rasterio 1.5.2
Hardware	Google Colab free tier, NVIDIA Tesla T4 (15 GB)

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